

Calculus Skeleton Notes

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Average Rates of Change

Goals: Compute average rates of change, slopes of secant lines, and motivate the idea of limits

Before Class Videos: None today, starting next class

Example 1: I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

b) Sketch a possible graph

c) How could I approximate my speed at $t = 10$ minutes?

Example 2: What is average rate of change of $f(x) = x^2$ on $[2, 4]$?

Example 3: Find the equation of the line through (x_1, y_1) and (x_2, y_2) .

Example 4: Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

Big Idea: As x_2 gets closer to x_1 the slope of the secant line approaches the slope of the tangent line at x_1 . We need to investigate and define this limiting process more precisely. This DESMOS visualizes the idea: <https://www.desmos.com/calculator/oi7poyhodr>

Suggested after class problems from APEX: None today, but review the Course Outline on Brightspace

1.1 - Limits

Goals: Describe limits graphically, numerically, and algebraically.

Before Class Videos: 1.1.1 and 1.1.2

Example 1: Consider these three functions:

$$f(x) = x + 1, \quad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x + 1)(x - 1)}{x - 1}, \quad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?

Definition:

$\lim_{x \rightarrow a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a .

Example 2: Compute the limit as $x \rightarrow 1$ of $f(x)$, $g(x)$, and $h(x)$.

Example 3: Use a table of values to estimate $\lim_{x \rightarrow 1} (x + 1)$.

Example 4: Use a table of values to estimate $\lim_{x \rightarrow 0} \frac{1}{x}$.

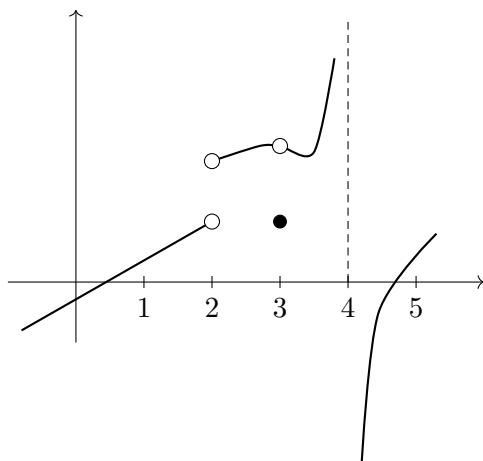
Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

Definition:

$\lim_{x \rightarrow a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right,
i.e. $x > a$.

Example 6: Use the graph to evaluate the indicated limits.



$$\lim_{x \rightarrow 2^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 2^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 2} f(x) = \underline{\hspace{2cm}},$$

$$\lim_{x \rightarrow 3^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 3^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 3} f(x) = \underline{\hspace{2cm}},$$

$$\lim_{x \rightarrow 4^-} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 4^+} f(x) = \underline{\hspace{2cm}}, \quad \lim_{x \rightarrow 4} f(x) = \underline{\hspace{2cm}}.$$

Example 7: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$

Example 8: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

Suggested after class problems from APEX: Section 1.1 problems 9,11,15

1.3 - Finding Limits Analytically

Goals: Evaluate limits using limit laws, algebra, and the Squeeze Theorem.

Before Class Videos: 1.3.1–1.3.3

Definition:

If $f(x)$ gets arbitrarily close to L for all x sufficiently close to a , then

$$\lim_{x \rightarrow a} f(x) = L.$$

For many “nice” functions, evaluating a limit just means plugging in the number. We’ll investigate what functions are “nice” in section 1.5 on Continuity. For example:

Example 1:

- $\lim_{x \rightarrow 2} x^3 =$
- $\lim_{x \rightarrow \pi} \sin(x) =$
- $\lim_{x \rightarrow e} \ln(x) =$
- $\lim_{x \rightarrow 4} \sqrt{x} =$

Theorem:

If L , M , c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M,$$

then

- | | |
|----------------------------|---|
| 1. Sum Rule: | $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$ |
| 2. Difference Rule: | $\lim_{x \rightarrow c} [f(x) - g(x)] = L - M$ |
| 3. Constant Multiple Rule: | $\lim_{x \rightarrow c} [kf(x)] = kL$ |
| 4. Product Rule: | $\lim_{x \rightarrow c} [f(x)g(x)] = LM$ |
| 5. Quotient Rule: | $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$ |
| 6. Power Rule: | $\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \text{ a positive integer}$ |
| 7. Root Rule: | $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$ |

If n is even in the Root Rule, assume that $f(x) \geq 0$ on an interval containing c .

Example 2: Evaluate

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x + 2}.$$

Example 3: Come up with functions f and g such that

$$\lim_{x \rightarrow 3} f(x)g(x) \neq \left[\lim_{x \rightarrow 3} f(x) \right] \left[\lim_{x \rightarrow 3} g(x) \right].$$

Four Algebraic Tricks

Example 4:

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x - 2}.$$

Example 5:

$$\lim_{x \rightarrow 0} \frac{(x - 2)^2 - 4}{x}.$$

Example 6:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}.$$

Example 7:

$$\lim_{x \rightarrow 2} \frac{\frac{1}{2} - \frac{1}{x}}{x - 2}.$$

Example 8: Evaluate

$$\lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}}.$$

Theorem:

Squeeze Theorem. If $f(x) \leq g(x) \leq h(x)$ for all x in an interval containing a (except possibly at a), and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} g(x) = L.$$

Example 9: Evaluate

$$\lim_{x \rightarrow 0} x^2 \sin(1/x^2).$$

Example 10: Suppose

$$-x^2 + 4x + 1 \leq g(x) \leq -2x + 10$$

for all x . For what value of a can you determine $\lim_{x \rightarrow a} g(x)$? What is the limit?